Muirfield High School

YEAR 9 - MATHEMATICS

Topic Test: Working Mathematically (Time Allowed: 30 minutes)

Name: ---

Class: ---------------------   Mark: -------------/ 20   **(Show all working out)**

1. John is 3 years older than his sister Amber. The product of their ages is the same as their father's age. If John's father is 40 years of age, how old is Amber?

2. Students in Mr. Finch's class had a total of \$360.00 to spend at the school fete. Each student had on average \$14.40. How many students are in Mr. Finch's class?

3. There are 26 students in Mrs Block's mathematics class: 14 girls and 12 boys. A total of 6 students were late to class. If 5 of the students late to class were boys, what is the ratio of Girls to Boys that were on time?

Using a Venn diagram

4. In a class of 22 students, each attends swimming practice, or lifesaving, or both. If 16 attend swimming practice and 9 attend lifesaving, find the number of students who attend both.

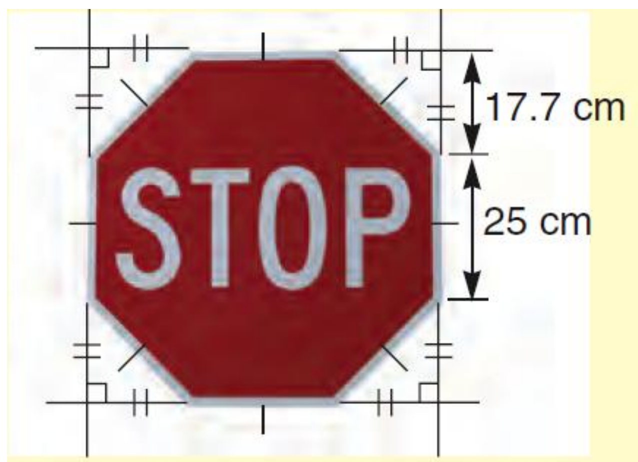
5. The council rates on my block of land are calculated at the rate of 0.83 cents for every \$1 on the value of my land. If my land is valued at \$145 000, how much must I pay to the nearest dollar?

6. A chocolate bar has 40 pieces. If Mary and Sue shared the pieces in the ratio 3 : 5, how many pieces did each get?

7. Packets of food are checked for weight. A machine records the following:
Overweight: 643
Underweight: 490
Correct weight: 24 608

What percentage of the packets weighed were underweight? Answer to the nearest whole percent.

8. Calculate the area of metal in the 'STOP' sign.



9. Which two numbers have a sum of 1200 and a difference of 68?

10. A car is travelling at an average speed of 80 km/h. At this rate find:
    a. how far it will travel in 6 h
    b. how long it would take to travel 980km

THE END

Muirfield High School

YEAR 9 - MATHEMATICS

Topic Test: Working Mathematically (Time Allowed: 30 minutes)

Name: -<u>SOLUTIONS</u>-

Class: --------------------- Mark: -------------/ 20 (Show all working out)

1. John is 3 years older than his sister Amber. The product of their ages is the same as their father's age. If John's father is 40 years of age, how old is Amber?

   Trial and Error

   $$\left.\begin{array}{lll} \mathbf{4 \times 10 = 40} & \mathbf{10 - 4 = 6} & \times \\ \mathbf{2 \times 20 = 40} & \mathbf{20 - 2 = 18} & \times \\ \mathbf{8 \times 5 = 40} & \mathbf{8 - 5 = 3} & \checkmark \end{array}\right\} \checkmark$$

   1 mark for some working out.

   Amber's age is 5 year. ✓

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2. Students in Mr. Finch's class had a total of \$360.00 to spend at the school fete. Each student had on average \$14.40. How many students are in Mr. Finch's class?

   $\mathbf{360.00 \div 14.40 = 25}$ ✓

   There are 25 students in Mr Finch's class. ✓

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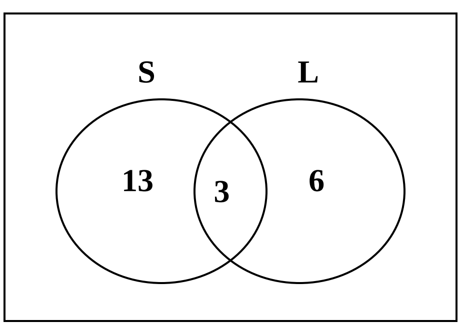
3. There are 26 students in Mrs Block's mathematics class: 14 girls and 12 boys. A total of 6 students were late to class. If 5 of the students late to class were boys, what is the ratio of Girls to Boys that were on time?

   G:B = (14 – 1) : (12 – 5)

   13 : 7 were on time.
   ✓ ✓

Using a Venn diagram

4. In a class of 22 students, each attends swimming practice, or lifesaving, or both. If 16 attend swimming practice and 9 attend lifesaving, find the number of students who attend both.



✓

16 + 9 – 22 = 3

3 students attend both Swimming and Life Saving practice. ✓

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5. The council rates on my block of land are calculated at the rate of 0.83 cents for every \$1 on the value of my land. If my land is valued at \$145 000, how much must I pay to the nearest dollar?

   \$145 000 x 0.83 = 120 350 cents = \$1203.50 ≑ \$1204
   ✓ ✓

6. A chocolate bar has 40 pieces. If Mary and Sue shared the pieces in the ratio 3 : 5, how many pieces did each get?

Mary got $\frac{3}{8} \times 40 = 15$ **pieces.** ✓

Sue got $\frac{5}{8} \times 40 = 25$ **pieces.** ✓

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7. Packets of food are checked for weight. A machine records the following:
Overweight: 643
Underweight: 490
Correct weight: 24 608

What percentage of the packets weighed were underweight? Answer to the nearest whole percent.

Total = 25741 Underweight% = $\frac{490}{25741} \times 100$**=1.9035%** ✓

$\approx 2\%$ ✓

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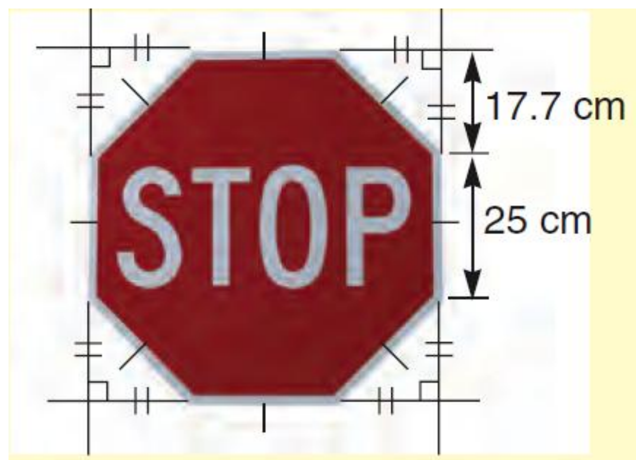
8. Calculate the area of metal in the 'STOP' sign.



Area of Metal = Area of Sq. – Area of 4 triangles

$= 60.4 \times 60.4 - 4 \times \left(\frac{1}{2} \times 17.7 \times 17.7\right)$

✓

$= 3021.58\ cm^2$ ✓

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9. Which two numbers have a sum of 1200 and a difference of 68?

Trial and Error

$$\left.\begin{array}{lll} 638 - 570 = 68 & 638 + 570 = 1208 & \times \\ 635 - 567 = 68 & 635 + 567 = 1202 & \times \\ 634 - 566 = 68 & 634 + 566 = 1200 & ✓ \end{array}\right\}$$

✓ **1 mark for some working out.**

The numbers are 634 and 566. ✓

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10. A car is travelling at an average speed of 80 km/h. At this rate find:
   a. how far it will travel in 6 h
   b. how long it would take to travel 980km

(a) $80 \times 6 = 480\ km$ ✓

(b) $\frac{980}{80} = 12.25\ h\ or\ 12h\ 15\ min.$ ✓

THE END